Wireless Networking

1. List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.

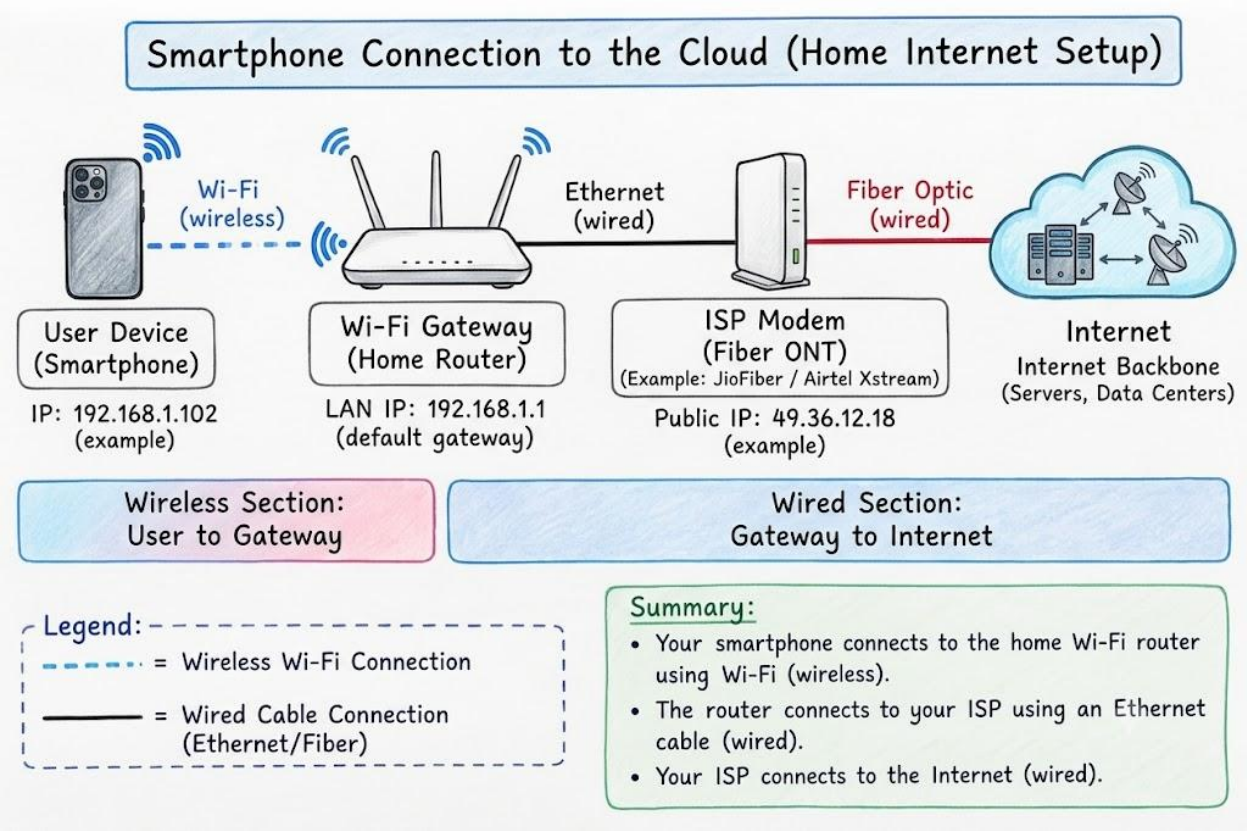
Ans:-

Standard	Wi-Fi Name	Frequency Band	Max Speed (Theory)	Key Difference
802.11n	Wi-Fi 4	2.4 GHz and 5 GHz	Up to 600 Mbps	Supports both frequency bands and uses MIMO technology.
802.11ac	Wi-Fi 5	5 GHz	Up to 6.9 Gbps	Offers higher speeds and wider channels than Wi-Fi 4.
802.11ax	Wi-Fi 6	2.4 GHz and 5 GHz	Up to 9.6 Gbps	Handles multiple connected devices more efficiently.

Note: These are theoretical maximum speeds. Actual speeds are usually lower. Wi-Fi 6E also supports the 6 GHz band.

2. Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.

Ans:-



- **Wireless part:** the smartphone connects to the Wi-Fi router using radio waves (2.4 GHz or 5 GHz).

- **Wired part:** the router connects to the ISP through an Ethernet or fiber cable, and the ISP connects to the Internet.

3. Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why.

Hint: Consider both signal strength and security when choosing.

Ans:-

- **How to scan:** On a phone, open Settings > Wi-Fi and tap each network for details.

On a Windows laptop, open Command Prompt and run: netsh wlan show networks mode=bssid

This shows the SSID, signal (in %) and authentication type of each network nearby. On Linux, use nmcli dev wifi list.

Important: The table below is a SAMPLE. Replace it with the networks you actually see on your own device before submitting.

SSID (network name)	Signal strength	Security type	Chosen?
HomeWiFi_5G	-45 dBm (Excellent)	WPA3	Yes
Cafe_Free_WiFi	-55 dBm (Good)	Open (no security)	No
Neighbour_2.4G	-78 dBm (Weak)	WPA2	No

* **Signal strength guide:**

Signal strength (dBm)	Quality	Meaning
-30 to -50	Excellent	Very close to the router, fastest speed
-50 to -60	Good	Reliable for streaming and video calls
-60 to -70	Fair	Usable but can slow down
Below -70	Weak	Frequent drops and slow speed

My choice: I would choose HomeWiFi_5G (sample). It has the strongest signal (-45 dBm), so the speed will be fast and stable, and it uses WPA3, which is the newest and most secure type. I would not choose Cafe_Free_WiFi because it is open, so other people nearby can capture my data. I would also avoid Neighbour_2.4G because its signal is weak and it is not my own network. Signal strength and security are both important, and the best choice is the one with a strong signal and strong security.

4. Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

Ans:-

What it is: The SSID is the name of a Wi-Fi network. Normally the router keeps announcing this name so that phones and laptops can see it in the list of networks. In SSID hiding, the router stops announcing the name, so the network does not appear in the list. To connect, the user must type the exact network name and the password by hand.

- **Advantage:** The network is not visible to casual users or neighbours, so fewer people try to connect to it, and the list of networks stays less crowded.

- **Disadvantage:** It is not real security. The name can still be found with free tools because the hidden network name is revealed when devices connect to it. It is also inconvenient, since every device needs the name typed in manually, and phones keep searching for the hidden network, which can use more battery.

Conclusion: Hiding the SSID can be an extra step, but the real protection comes from using a strong password with WPA2 or WPA3.